IES | GENAI FOR ENGINEERS

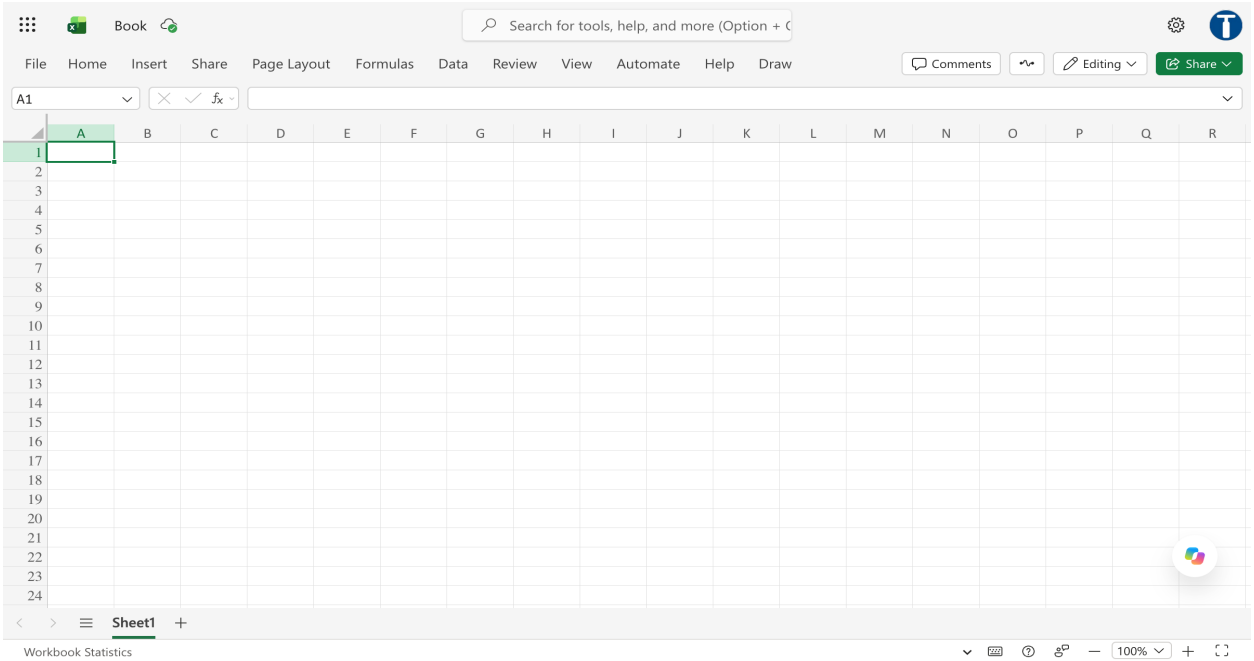
Lab 2 - Use Copilot in Excel to create a DMAIC summary

20 minutes | Synthetic classroom case

The team compares four baseline weeks with a two-week pilot after a maintenance change. All values are synthetic training data.

1	Open Starter.xlsx and select a cell in WeeklyData.
2	Open Copilot in Excel.
3	Copy the prompt from the Copilot_Prompt sheet.
4	Ask Copilot to create the output and show its formulas.
5	Check one result yourself; correct the draft if needed.

Where to find Copilot



Build the prompt with RISE

Role - Input - Steps - Expectations

R - Role	Six Sigma project assistant
I - Input	Only WeeklyData; four Baseline and two Pilot weeks
S - Steps	Calculate both rates; create five-row DMAIC summary
E - Expectations	Show formulas; separate facts from hypotheses; no causal claim

Paste the complete prompt into Copilot in Excel

Act as a Six Sigma project assistant. Use only the named Excel table WeeklyData. Calculate total opportunities, total deviations, and deviation rate for Baseline and Pilot using deviations divided by opportunities. Create a five-row DMAIC summary: Define, Measure, Analyse, Improve, Control. Clearly separate observed facts from hypotheses. State that the short pilot does not prove the cause of the change. Show the formulas and return one next evidence check.

Check one result yourself

Verify Baseline = 36 / 400 = 9.0% and Pilot = 7 / 200 = 3.5%. The lower pilot rate is an observation, not causal proof.

Expected Copilot draft

Two formula-based rates and a concise DMAIC table that states the problem, evidence, proposed test, and monitoring rule.

Microsoft 365 Copilot links

Microsoft 365 Copilot

Get started with Copilot in Excel

How Copilot Chat works in Microsoft 365 apps